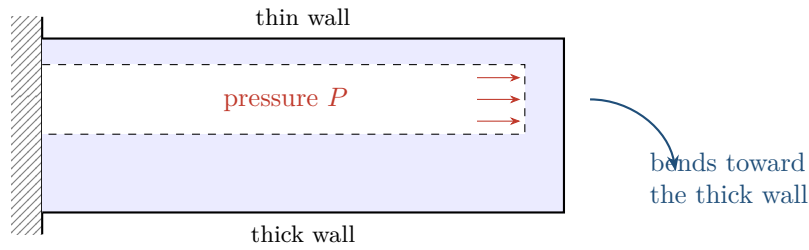


B.Tech Project

# Pressure-to-Deflection Modelling of an Eccentric-Cavity Soft Pneumatic Bending Actuator

Revised analytical derivation, corrections to the working notes,  
and verification by nonlinear finite-element analysis



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**Abstract.** A silicone cylinder with an off-centre (eccentric) cavity bends when the cavity is pressurised. This document restates, in a clean and corrected form, the derivation of the pressure-to-deflection relation given in the handwritten working notes (BTP\_260825\_144004.pdf). The cross-section properties, the eccentric thrust moment and the Euler–Bernoulli response are derived step by step. Typographical and sign errors are corrected, and the validity of each result is stated. Two physical corrections, both confirmed by a verified 3-D nonlinear finite-element (FE) model, change how the result is read. First, for a bare incompressible actuator the beam formula is multiplied by  $(1 - 2\nu)$ , which is zero for silicone. Second, the bending a bare actuator does show is a finite-strain ballooning effect that is quadratic in  $P$  at low pressure and ends at a limit pressure. The beam formula of the notes is the correct first-order model for a *hoop-reinforced* actuator.

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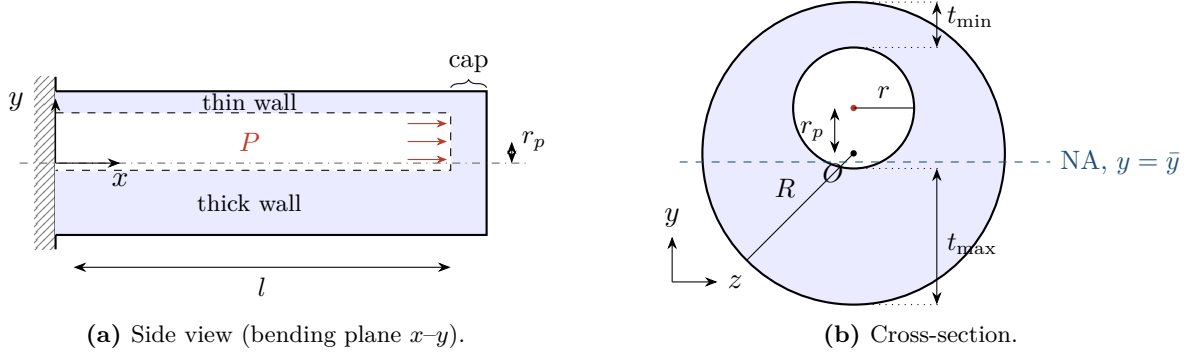
**How to read this document** Each section follows the order of the handwritten notes. Wherever the notes need a change, a red **Correction** box gives the original, the corrected form and the reason. Blue boxes are remarks. All corrections are collected in Table 6 (Section 10). Units throughout: mm, N, MPa ( $1 \text{ MPa} = 1000 \text{ kPa} = 1 \text{ N/mm}^2$ ).

# 1 Introduction

## 1.1 The device

The actuator is a solid silicone cylinder of outer radius  $R$  and pressurised length  $l$ . A circular cavity of radius  $r$  runs along it, with its centre offset by  $r_p$  from the axis of the cylinder. One wall is therefore thin,  $t_{\min} = R - r_p - r$ , and the opposite wall is thick,  $t_{\max} = R + r_p - r$ . The base is clamped, the far end is sealed by a solid cap, and the cavity is pressurised to a gauge pressure  $P$ . The actuator curls toward the thick wall (Figure 1).

**Objective.** Find the map from the input  $P$  to the tip bending angle  $\theta$  and the tip deflection  $\delta$ .



**Figure 1:** Geometry and coordinate frame. The origin  $O$  is the centre of the outer circle, the cavity centre is at  $(y, z) = (r_p, 0)$  and the neutral axis (NA) lies at  $y = \bar{y} < 0$ . Not to scale.

## 1.2 Scope

Sections 3–5 reproduce the beam derivation of the notes (eqs. 1–7) with corrections. Section 6 adds the effect of the in-plane wall stresses, which the notes leave out and which decides whether a bare actuator bends at first order. Section 7 revises the “asymmetric wall expansion” analysis of pages 5–7 of the notes. Section 8 lists the model hierarchy and Section 9 summarises the FE verification. All numerical values refer to the *placeholder* design  $R = 10$ ,  $r = 4$ ,  $r_p = 3$ ,  $l = 100$  mm with a neo-Hookean silicone of  $E = 0.6$  MPa.

# 2 Notation and assumptions

| Symbol                   | Meaning                                                       | Placeholder value               |
|--------------------------|---------------------------------------------------------------|---------------------------------|
| $R$                      | outer radius                                                  | 10 mm                           |
| $r$                      | cavity radius                                                 | 4 mm                            |
| $r_p$                    | cavity eccentricity (offset of the cavity centre along $+y$ ) | 3 mm                            |
| $l$                      | pressurised length                                            | 100 mm                          |
| $t_{\min}, t_{\max}$     | thinnest and thickest wall, $R \mp r_p - r$                   | 3 mm, 9 mm                      |
| $E, \mu, \nu$            | small-strain Young’s modulus, shear modulus, Poisson’s ratio  | 0.6 MPa, 0.2 MPa, $\approx 0.5$ |
| $C_{10}, C_{20}, C_{30}$ | Yeoh coefficients (neo-Hookean: $C_{10} = \mu/2 = E/6$ only)  | $C_{10} = 0.1$ MPa              |
| $P$                      | gauge pressure in the cavity (the input)                      | 0 kPa to 110 kPa                |
| $A, A_c$                 | elastomer area, cavity area $\pi r^2$                         |                                 |
| $\bar{y}, I_{zz}, e$     | neutral-axis position, second moment about it, moment arm     |                                 |
| $\kappa, \theta, \delta$ | curvature, tip angle, tip deflection                          |                                 |

**Table 1:** Symbols. Axes:  $x$  along the actuator (base at  $x = 0$ , tip at  $x = l$ );  $y$  in the section, from the outer centre toward the cavity centre;  $z$  completes the right-handed frame. Bending is about  $z$ , toward  $-y$ .

**Assumptions.**

- (A1) The material is isotropic, homogeneous, rate-independent and (nearly) incompressible; Mullins and viscoelastic effects are removed by preconditioning.
- (A2) The pressure is uniform in the cavity; gravity and external loads are neglected; the base is clamped and the tip is free.
- (A3) *Beam model only*: plane sections remain plane, the cross-section keeps its shape, and every longitudinal fibre is in uniaxial stress. Assumption (A3) holds for a hoop-reinforced actuator and fails for a bare one (Section 6).

### 3 Cross-section geometry

#### 3.1 Elastomer area

The section is the outer disc minus the cavity disc:

$$A = \pi R^2 - \pi r^2 = \pi (R^2 - r^2). \quad (3.1)$$

#### 3.2 Neutral axis

Treat the cavity as a negative area centred at  $y = r_p$ . The centroid of the composite area is

$$\bar{y} = \frac{\pi R^2 \cdot 0 - \pi r^2 \cdot r_p}{\pi R^2 - \pi r^2} \implies \boxed{\bar{y} = -\frac{r^2 r_p}{R^2 - r^2}} \quad (3.2)$$

The neutral axis sits on the thick-wall side of the outer centre ( $\bar{y} < 0$ ), because material has been removed on the  $+y$  side.

**Correction to the notes (notes eq. 2)** The notes write the denominator as  $R^2 - r$ . It must be  $R^2 - r^2$ ; the line above it in the notes already has  $\pi(R^2 - r^2)$ , so this is a transcription slip. Dimensional check:  $r^2 r_p / (R^2 - r^2)$  is a length,  $r^2 r_p / (R^2 - r)$  is not.

#### 3.3 Second moment of area about the neutral axis

By the parallel-axis theorem  $I = I_c + Ad^2$  applied to each disc, with distances measured from the neutral axis:

$$I_{zz} = \left[ \frac{\pi R^4}{4} + \pi R^2 \bar{y}^2 \right] - \left[ \frac{\pi r^4}{4} + \pi r^2 (r_p - \bar{y})^2 \right]. \quad (3.3)$$

To simplify, note that  $r_p - \bar{y} = r_p + \frac{r^2 r_p}{R^2 - r^2} = \frac{R^2 r_p}{R^2 - r^2}$ . Then

$$\pi R^2 \bar{y}^2 - \pi r^2 (r_p - \bar{y})^2 = \frac{\pi r_p^2}{(R^2 - r^2)^2} [R^2 r^4 - r^2 R^4] = -\frac{\pi R^2 r^2 r_p^2 (R^2 - r^2)}{(R^2 - r^2)^2} = -\frac{\pi R^2 r^2 r_p^2}{R^2 - r^2}, \quad (3.4)$$

so

$$\boxed{I_{zz} = \frac{\pi (R^4 - r^4)}{4} - \frac{\pi R^2 r^2 r_p^2}{R^2 - r^2}} \quad (3.5)$$

The first term is the concentric hollow cylinder; the second is the stiffness lost by shifting the hole off-centre. This agrees with eq. 3 of the notes. For the placeholder design  $I_{zz} = \pi(2436 - 171.43) = 7114.4 \text{ mm}^4$  (the parallel-axis form evaluated numerically gives the same value).

## 4 Pressure loading

### 4.1 Thrust on the sealed tip

The pressure acts on the inside face of the tip cap, whose projected area is the cavity area  $A_c = \pi r^2$ :

$$F = P A_c = P \pi r^2. \quad (4.1)$$

**Correction to the notes** (notes eq. 4, notation) The notes write  $F = PA$ . The symbol  $A$  is already the elastomer area of eq. (3.1), and using it here invites the error  $F = P\pi(R^2 - r^2)$ . Write  $F = PA_c$  with  $A_c = \pi r^2$ .

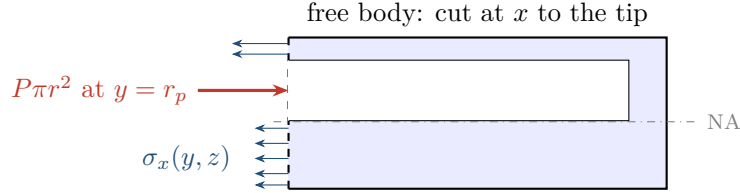
### 4.2 Moment arm

The resultant of a uniform pressure on a circular face passes through the centre of the circle,  $y = r_p$ . Its distance from the neutral axis is

$$e = r_p - \bar{y} = r_p + \frac{r^2 r_p}{R^2 - r^2} \implies e = \frac{R^2 r_p}{R^2 - r^2} \quad (4.2)$$

### 4.3 Bending moment

$$M = F e = \frac{\pi P R^2 r^2 r_p}{R^2 - r^2} \quad (4.3)$$



**Figure 2:** Free body from a cut at  $x$  to the tip, silicone together with the fluid it encloses. The fluid beyond the cut pushes with  $P\pi r^2$  through the cavity centre; the silicone carries the axial stress  $\sigma_x$ .

**Why  $M$  is uniform along the length** Cut the actuator at any  $x$  and take the free body from the cut to the tip, *including the fluid inside it* (Figure 2). No external force acts on it except at the cut. Equilibrium gives

$$\int_A \sigma_x dA = P\pi r^2, \quad \int_A \sigma_x (y - r_p) dA = 0. \quad (4.4)$$

So the wall is in net tension  $N = P\pi r^2$  and the stress resultant acts through the cavity centre. About the neutral axis this is the moment  $M = Ne$  of eq. (4.3), at every  $x$  and for any material law. Eq. (4.4) fixes the *resultants* of  $\sigma_x$  only; turning them into strain needs a constitutive assumption (Sections 5 and 6).

## 5 Beam-theory response (Euler–Bernoulli)

Under assumption (A3),  $\sigma_x = E \varepsilon_x$  with  $\varepsilon_x = \varepsilon_0 + \kappa (y - \bar{y})$ .

## 5.1 Curvature

$$\kappa = \frac{1}{\rho} = \frac{M}{EI_{zz}} = \frac{\pi P R^2 r_p^2}{E I_{zz} (R^2 - r^2)} \quad (5.1)$$

**Correction to the notes (notes p. 3, meaning of  $E$ )** The notes allow “Young’s modulus or effective secant modulus”. Use the small-strain modulus,  $E = 3\mu = 6C_{10}$  for an incompressible solid. A secant modulus chosen to fit a curve hides the real nonlinearity, which for the bare actuator is ballooning (Section 7), not a softening of  $E$ .

## 5.2 Tip bending angle

$M$  is uniform, so  $\kappa$  is uniform and the centreline is a circular arc:

$$\theta = \kappa l = \frac{M l}{EI_{zz}} = \frac{\pi P R^2 r_p^2 l}{E I_{zz} (R^2 - r^2)} \quad (5.2)$$

**Correction to the notes (notes p. 3, tip angle)** The notes write  $P^2$  in the last expression for  $\theta$ . It must be  $P$ :  $\theta = \kappa l$  and  $\kappa$  is linear in  $P$ . With  $P^2$  the units do not balance either.

## 5.3 Tip deflection

For an arc of radius  $\rho = 1/\kappa$  turning through  $\theta$ , the tip moves sideways by

$$\delta = \rho (1 - \cos \theta) = \frac{1 - \cos \theta}{\kappa} = \frac{l}{\theta} (1 - \cos \theta). \quad (5.3)$$

For  $\theta \ll 1$ ,  $1 - \cos \theta \approx \theta^2/2$  and

$$\delta \approx \frac{\theta l}{2} = \frac{M l^2}{2 E I_{zz}}, \quad (5.4)$$

which is the cantilever result for a uniform end moment. Both forms agree with page 4 of the notes. The axial tip position is  $x_{\text{tip}} = \rho \sin \theta$ .

## 5.4 Axial elongation

The net tension of eq. (4.4) stretches the centroidal fibre by  $\varepsilon_0 = F/(EA)$ , so  $\Delta l = Fl/(EA)$ .

**Correction to the notes (notes p. 4, note b)** (i) A uniform axial strain moves every fibre by the same amount and does *not* shift the neutral axis; the remark should be dropped. (ii)  $\Delta l = Fl/(EA)$  holds only under assumption (A3), i.e. for a hoop-reinforced actuator. For a bare incompressible actuator the first-order elongation is  $(1 - 2\nu)Fl/(EA) \approx 0$  (Section 6).

## 5.5 Non-dimensional form

Writing  $\hat{I} = I_{zz}/R^4$ ,

$$\theta = \frac{P}{E} \frac{l}{R} f\left(\frac{r}{R}, \frac{r_p}{R}\right), \quad f = \frac{\pi (r/R)^2 (r_p/R)}{\hat{I} (1 - (r/R)^2)}. \quad (5.5)$$

The response is linear in  $P/E$  and in the slenderness  $l/R$ . The section shape enters only through  $f$ . For the placeholder design  $\hat{I} = 0.7114$  and  $f = 0.2523$ .

## 5.6 Worked example

| Section (eqs. 3.1–4.2) |                        | Response at $P = 20$ kPa (eqs. 4.1–5.4) |                                        |
|------------------------|------------------------|-----------------------------------------|----------------------------------------|
| $A$                    | 263.9 mm <sup>2</sup>  | $F$                                     | 1.005 N                                |
| $\bar{y}$              | −0.571 mm              | $M$                                     | 3.590 N mm                             |
| $I_{zz}$               | 7114.4 mm <sup>4</sup> | $\kappa$                                | $8.41 \times 10^{-4}$ mm <sup>−1</sup> |
| $e$                    | 3.571 mm               | $\theta$                                | 4.82°                                  |
| $t_{\min}, t_{\max}$   | 3 mm, 9 mm             | $\delta$ arc / small-angle              | 4.203 mm / 4.206 mm                    |

**Table 2:** Beam model for the placeholder design,  $E = 0.6$  MPa. At 10 kPa, 40 kPa and 60 kPa:  $\theta = 2.41^\circ$ ,  $9.64^\circ$  and  $14.46^\circ$ ,  $\delta_{\text{arc}} = 2.10$  mm, 8.39 mm and 12.55 mm.

**Validity of eqs. (5.1)–(5.4).** They rest on assumption (A3): each fibre carries only axial stress. That is true when a hoop-fibre sleeve (thread wrap) carries the hoop load. The FE model confirms that for the *sleeved* actuator the tip angle matches eq. (5.2) within 5 % up to  $P/E \approx 0.03$ . For the *bare* actuator see the next section.

## 6 The bare actuator: effect of the in-plane wall stresses

The pressure does not only push on the tip cap. It also loads the wall *in the plane of the section*: radial compression at the cavity surface and hoop tension around it, both of order  $P$  or larger. Assumption (A3) ignores them. In linear isotropic elasticity they enter the axial strain through Poisson’s ratio:

$$\varepsilon_x = \frac{\sigma_x - \nu(\sigma_y + \sigma_z)}{E} \implies \sigma_x = E[\varepsilon_0 + \kappa(y - \bar{y})] + \nu(\sigma_y + \sigma_z). \quad (6.1)$$

### 6.1 Resultants of the in-plane stresses

The in-plane field need not be solved: its resultant and first moment follow from equilibrium alone. Far from the ends the in-plane stresses  $\sigma_{\alpha\beta}$  ( $\alpha, \beta \in \{y, z\}$ ) satisfy  $\partial_\beta \sigma_{\alpha\beta} = 0$ , the outer surface is traction-free and the cavity surface carries the traction  $P \mathbf{m}$ , where  $\mathbf{m}$  is the unit normal pointing out of the cavity into the wall. Take coordinates  $\mathbf{x}'$  centred on the cavity. From the identities

$$\partial_\beta(\sigma_{\alpha\beta} x'_\alpha) = \sigma_{\alpha\alpha}, \quad \partial_\beta(\sigma_{\alpha\beta} x'_\alpha x'_\gamma) - \partial_\beta(\sigma_{\gamma\beta} \tfrac{1}{2} |\mathbf{x}'|^2) = \sigma_{\alpha\alpha} x'_\gamma$$

and the divergence theorem, with  $\mathbf{x}' = r\mathbf{m}$  on the cavity boundary,

$$\int_A (\sigma_y + \sigma_z) dA = \oint P r ds = 2P\pi r^2, \quad \int_A (\sigma_y + \sigma_z) x'_\gamma dA = \oint P r^2 m_\gamma ds - \oint \frac{P r^2}{2} m_\gamma ds = 0. \quad (6.2)$$

So the in-plane stresses have a total of  $2P\pi r^2$  acting *through the cavity centre*, for an outer boundary of any shape. About the neutral axis:

$$\int_A (\sigma_y + \sigma_z) dA = 2P\pi r^2, \quad \int_A (\sigma_y + \sigma_z)(y - \bar{y}) dA = 2P\pi r^2 e. \quad (6.3)$$

### 6.2 Corrected first-order response

Substitute eq. (6.1) into the two free-body conditions (4.4), using  $\int_A (y - \bar{y}) dA = 0$ :

$$\text{force: } EA\varepsilon_0 + 2\nu P\pi r^2 = P\pi r^2, \quad (6.4)$$

$$\text{moment: } EI_{zz}\kappa + 2\nu P\pi r^2 e = P\pi r^2 e. \quad (6.5)$$

$$\varepsilon_0 = (1 - 2\nu) \frac{P\pi r^2}{EA}, \quad \kappa = (1 - 2\nu) \frac{M}{EI_{zz}} = (1 - 2\nu) \frac{\pi P R^2 r^2 r_p}{E I_{zz} (R^2 - r^2)}. \quad (6.6)$$

For silicone  $\nu \rightarrow 0.5$ : **in the small-strain limit a bare eccentric-cavity actuator neither lengthens nor bends.**

The first result is the textbook statement that a closed incompressible tube does not lengthen under internal pressure; the second is its bending counterpart. The thrust moment is still there, but the Poisson effect of the in-plane stresses cancels it exactly.

**Correction to the notes** (notes eqs. 4–7) State assumption (A3) with eqs. 4–7 and the case it holds for (hoop-reinforced). Add eq. (6.6) for the bare actuator.

**FE check of eq. (6.6)** Mid-span curvature from the FE model divided by eq. (5.1), at  $P/E = 10^{-5}$ :

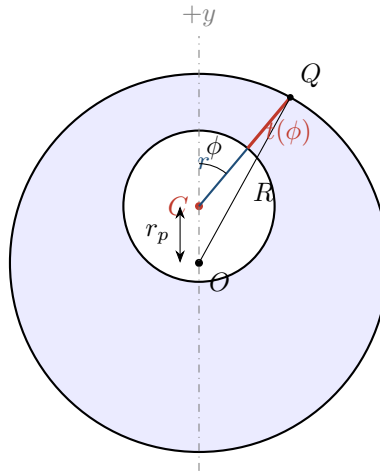
| $\nu$      | 0      | 0.2    | 0.3    | 0.4    | 0.45   | 0.4995 |
|------------|--------|--------|--------|--------|--------|--------|
| FE         | 1.0000 | 0.6001 | 0.4001 | 0.2002 | 0.1002 | 0.0012 |
| $1 - 2\nu$ | 1.0000 | 0.6000 | 0.4000 | 0.2000 | 0.1000 | 0.0010 |

The tip angle of the nearly incompressible actuator is not exactly zero but 6 % of eq. (5.2). All of it comes from the two ends, where the clamp and the cap stop the wall from expanding and switch the cancellation off locally.

## 7 Asymmetric wall expansion (notes pp. 5–7, revised)

The notes rightly identify a second mechanism. The thin wall stretches more in hoop than the thick wall, the stretch couples into axial lengthening, and the thin side becomes longer than the thick side. This section keeps the geometry of the notes, corrects the formulae and replaces the linear estimate by the correct finite-strain one.

### 7.1 Wall thickness around the cavity



**Figure 3:** Wall thickness along a ray from the cavity centre  $C$  at angle  $\phi$  from  $+y$ .  $\phi = 0$  points at the thin wall.

Along a ray from the cavity centre  $C$  at angle  $\phi$  from  $+y$  (Figure 3), the point  $Q$  at distance  $s = r + t$  lies on the outer circle. The law of cosines in triangle  $OCQ$  gives

$$R^2 = r_p^2 + s^2 + 2r_p s \cos \phi \implies s = -r_p \cos \phi + \sqrt{R^2 - r_p^2 \sin^2 \phi}, \quad (7.1)$$

taking the positive root, so

$$t(\phi) = \sqrt{R^2 - r_p^2 \sin^2 \phi} - r_p \cos \phi - r \quad (7.2)$$

with  $t(0) = R - r_p - r = t_{\min}$  and  $t(\pi) = R + r_p - r = t_{\max}$ . The FE mesh is built on this formula.

**Correction to the notes** (notes p. 5,  $t(\phi)$ ) The notes write  $\sin^2 \theta$  under the root and  $\cos \phi$  outside it. Use one angle,  $\phi$ , and state that it is measured at the cavity centre from the  $+y$  axis. The expression is exact for the radial wall thickness along that ray, so “ $\approx$ ” can be “ $=$ ”.

**Correction to the notes** (notes pp. 5–6, axis label) The notes call the longitudinal strain and stress  $\varepsilon_{zz}$  and  $\sigma_z$ , but in the frame of page 1 the actuator axis is  $x$  and  $z$  lies in the section. Write  $\varepsilon_x$  and  $\sigma_x$ .

## 7.2 Hoop stress

In the thin-membrane approximation the hoop stress at angle  $\phi$  is  $\sigma_\theta(\phi) \approx Pr/t(\phi)$ , and the mean axial stress is  $\bar{\sigma}_x = P\pi r^2/A$ . The membrane estimate needs  $t \ll r$ ; here  $t_{\min}/r = 0.75$ , so the values are indicative only.

## 7.3 The linear estimate of the notes, and why it cannot stand

The notes take  $\varepsilon_x(\phi) \propto Pr/(Et(\phi))$  and define

$$\kappa_{\exp} \approx \frac{\varepsilon_x(\text{thin}) - \varepsilon_x(\text{thick})}{2R} = \frac{Pr}{2ER} \left( \frac{1}{t_{\min}} - \frac{1}{t_{\max}} \right). \quad (7.3)$$

**Correction to the notes** (notes p. 6, sign) When eq. (7.3) is substituted into  $\kappa_{\text{total}}$  the bracket becomes  $(\frac{1}{R-r_p-r} + \frac{1}{R+r_p-r})$ . With the definition above it is  $(\frac{1}{R-r_p-r} - \frac{1}{R+r_p-r})$ . The sign error doubles the term for the placeholder design (4/9 instead of 2/9).

Even with the sign fixed, eq. (7.3) cannot be kept:

1. It rests on linear Poisson coupling, and Section 6 shows that for  $\nu = 0.5$  linear elasticity gives *no* axial strain and *no* curvature from pressure at all. The linear in-plane effect is exactly what cancels  $\kappa_{\text{thrust}}$ ; it cannot also add to it.
2. The superposition  $\kappa_{\text{total}} = \kappa_{\text{thrust}} + \kappa_{\exp}$  is linear in  $P$  and equals  $2.76 \kappa_{\text{thrust}}$  at every pressure for the placeholder design. The FE curvature of the bare actuator runs from 0.1 to 3 times  $\kappa_{\text{thrust}}$  over the pressure range: it is not linear in  $P$ .

## 7.4 The correct mechanism: second-order lengthening

Consider a thin closed neo-Hookean tube with hoop, axial and thickness stretches  $\lambda_1 = 1 + a$ ,  $\lambda_2 = 1 + b$ ,  $\lambda_3 = 1/(\lambda_1 \lambda_2)$ . The membrane stresses are  $\sigma_\theta = \mu(\lambda_1^2 - \lambda_3^2)$  and  $\sigma_x = \mu(\lambda_2^2 - \lambda_3^2)$ , and a closed end gives  $\sigma_x = \sigma_\theta/2$ :

$$2\lambda_2^2 = \lambda_1^2 + \lambda_3^2. \quad (7.4)$$

Expanding with  $\lambda_3^2 \approx 1 - 2a - 2b + 3a^2 + 3b^2 + 4ab$ :

- first order:  $4b = -2b$ , so  $b = 0$ . There is no axial strain at first order, in agreement with eq. (6.6);
- second order, with  $b = O(a^2)$ :  $6b = 4a^2$ , so

$$\varepsilon_{\text{axial}} \approx \frac{2}{3} \varepsilon_{\text{hoop}}^2 \quad (7.5)$$

The thin wall has the larger hoop strain, so it lengthens more and the actuator bends toward the thick wall. This is the mechanism the notes describe, but it is *quadratic* in  $P$ , not linear.

**Scaling estimate** Take the closed-tube membrane hoop strain  $\varepsilon_{\text{hoop}} = \frac{3}{4} Pr/(Et)$  for each wall, apply eq. (7.5), and divide the difference in axial strain by the distance between the wall mid-lines ( $d = 14 \text{ mm}$ ):

$$\kappa_{\text{wall}} \approx \frac{2}{3} \frac{\varepsilon_{\text{hoop}}(t_{\min})^2 - \varepsilon_{\text{hoop}}(t_{\max})^2}{d}.$$

This gives  $1.2 \times 10^{-5} \text{ mm}^{-1}$  at 10 kPa and  $4.7 \times 10^{-5} \text{ mm}^{-1}$  at 20 kPa, against FE mid-span values of  $1.8 \times 10^{-5}$  and  $7.8 \times 10^{-5} \text{ mm}^{-1}$ . The estimate is within a factor of two and reproduces the  $P^2$  scaling: the FE curvature grows by 4.3 when  $P$  doubles. It is a rough guide only, since the wall is not thin and the two walls interact.

## 7.5 Ballooning limit

A rubber tube under internal pressure has a pressure maximum: past it, the wall thins faster than the hoop stress can rise, and the tube inflates without bound at falling pressure. The thin wall reaches it first. For the placeholder shape the FE model gives the limit at

$$P_{\text{lim}} = 0.523 \mu \quad (104.7 \text{ kPa for } E = 0.6 \text{ MPa; } 14 \text{ kPa for } E = 0.08 \text{ MPa}). \quad (7.6)$$

Above  $P_{\text{lim}}$  a neo-Hookean actuator has no equilibrium under pressure control. Under volume control (a water-filled syringe) the path through the limit is stable. A real silicone stiffens at large stretch (Yeoh  $C_{20}, C_{30} > 0$ ), which can raise or remove the limit, so the material must be measured before this number is trusted.

**Correction to the notes** (notes p. 7, note) Keep the statement that wall compliance dominates. Add that for a bare actuator it is the only mechanism that survives at first order, and that it ends in a limit pressure.

## 8 Model hierarchy

| Tier | Content                                                                                                                                                                                                                                                                                                         | Applies to                                   |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| 1    | Linear Euler–Bernoulli, eqs. (5.1)–(5.4)                                                                                                                                                                                                                                                                        | hoop-reinforced, $P/E \lesssim 0.03$         |
| 2    | Tier 1 curvature with large-rotation arc kinematics and axial stretch: $l_c = l(1 + \varepsilon_0)$ ,<br>$\rho = l_c/\theta$ , $x_{\text{tip}} = \rho \sin \theta$ , $\delta = \rho(1 - \cos \theta)$                                                                                                           | hoop-reinforced; the form needed for control |
| 3    | Hyperelastic section equilibrium:<br>$\lambda(y) = \lambda_c + k(y - r_p)$ ; solve $\int s(\lambda) dA = P\pi r^2$ ,<br>$\int s(\lambda)(y - r_p) dA = 0$ with $s(\lambda) = 2(\lambda - \lambda^{-2})[C_{10} + 2C_{20}(I_1 - 3) + 3C_{30}(I_1 - 3)^2]$ ,<br>$I_1 = \lambda^2 + 2/\lambda$ ; then $\theta = kl$ | hoop-reinforced, larger $P$                  |
| 4    | Nonlinear 3-D finite elements (Section 9)                                                                                                                                                                                                                                                                       | bare and reinforced, up to the limit point   |
| 5    | Experiment                                                                                                                                                                                                                                                                                                      | material constants and validation            |

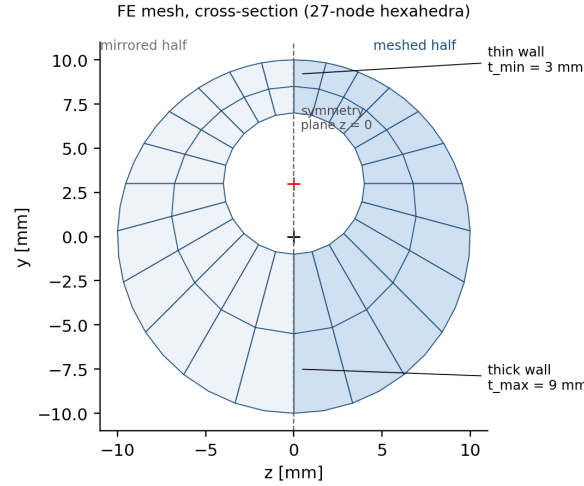
**Table 3:** Model tiers. Tiers 1–3 share assumption (A3) and describe the hoop-reinforced actuator; the bare actuator needs Tier 4.

## 9 Verification by nonlinear finite-element analysis

### 9.1 What was done and how

No commercial structural FE package was available, so a solver was written in Python with NumPy, SciPy and JAX (`actuator_fea.py`):

- **Mesh:** 27-node hexahedra, half model with symmetry about the bending plane (Figure 4); 588 elements (medium mesh); clamped base, solid tip cap of thickness  $R/2$ .
- **Material:** Yeoh/neo-Hookean with an isochoric–volumetric split, mixed displacement–pressure (Q2/P1) formulation to avoid volumetric locking,  $\nu = 0.4995$ .
- **Load:** follower pressure on every wetted cavity face, derived from the potential  $-PV$  of the enclosed cavity volume.
- **Solver:** Newton iteration with tangents from automatic differentiation; pressure control, switching automatically to volume control when a pressure step has no solution, so the limit point is passed and located.
- **Variants:** bare actuator, and the same actuator with an ideal inextensible hoop-fibre sleeve on the outer surface.



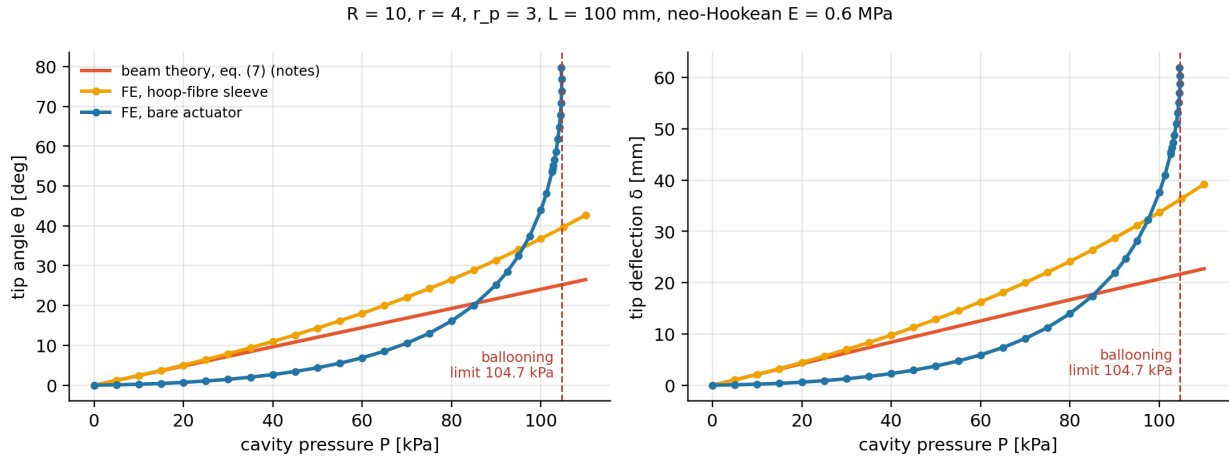
**Figure 4:** FE mesh of the cross-section. Only the half  $z \geq 0$  is solved; the other half follows by symmetry.

### 9.2 Verification of the solver

| Test                                              | Reference                        | Largest deviation                              |
|---------------------------------------------------|----------------------------------|------------------------------------------------|
| Uniaxial tension, $\lambda \leq 3$                | exact incompressible stress      | 0.2 % neo-Hookean; 2.5 % Yeoh at $\lambda = 3$ |
| Thick-walled tube inflation, $P/\mu \leq 0.7$     | exact neo-Hookean solution       | 0.05 %                                         |
| Small-pressure bending, $0 \leq \nu \leq 0.4995$  | $\kappa = (1 - 2\nu)M/(EI_{zz})$ | 0.0002 (in units of eq. 5.1)                   |
| Mesh refinement, $\delta$ at $P/\mu = 0.30, 0.45$ | fine mesh, 1664 elements         | medium mesh within 0.5 %                       |

**Table 4:** Verification (`fea_verify.py`, log in `results/verification.log`).

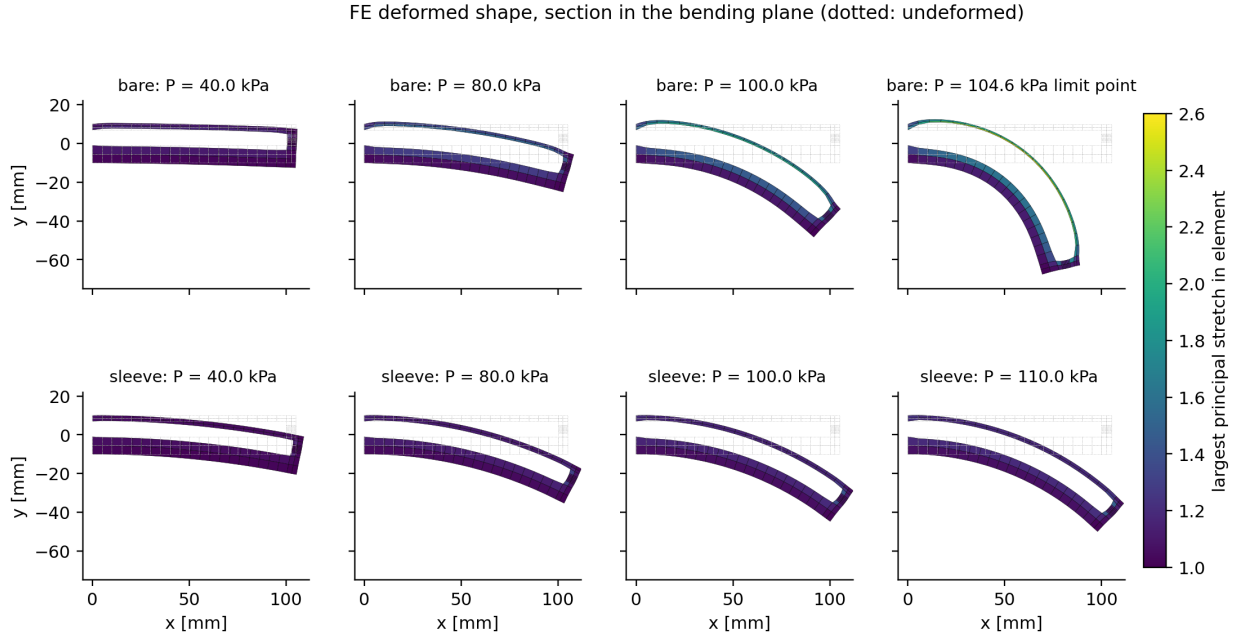
### 9.3 Findings



**Figure 5:** Tip angle and tip deflection against pressure: beam theory (eq. 5.2, 5.3) and FE for the bare and the sleeved actuator.

| $P$ [kPa] | $\theta_{\text{beam}}$ [°] | Bare         |         |               |         | Hoop-reinforced |         |               |         |
|-----------|----------------------------|--------------|---------|---------------|---------|-----------------|---------|---------------|---------|
|           |                            | $\theta$ [°] | FE/beam | $\delta$ [mm] | $V/V_0$ | $\theta$ [°]    | FE/beam | $\delta$ [mm] | $V/V_0$ |
| 10        | 2.41                       | 0.24         | 0.10    | 0.20          | 1.07    | 2.41            | 1.00    | 2.10          | 1.04    |
| 20        | 4.82                       | 0.71         | 0.15    | 0.59          | 1.14    | 5.03            | 1.04    | 4.42          | 1.09    |
| 40        | 9.64                       | 2.66         | 0.28    | 2.26          | 1.33    | 11.0            | 1.14    | 9.77          | 1.19    |
| 60        | 14.5                       | 6.89         | 0.48    | 5.91          | 1.59    | 18.1            | 1.25    | 16.3          | 1.30    |
| 80        | 19.3                       | 16.1         | 0.84    | 14.0          | 2.01    | 26.5            | 1.38    | 24.2          | 1.44    |
| 100       | 24.1                       | 44.0         | 1.82    | 37.6          | 2.98    | 36.8            | 1.53    | 33.7          | 1.61    |
| 104.65    | 25.2                       | 77           | 3.0     | 60            | 4.05    | —               | —       | —             | —       |

**Table 5:** FE results,  $E = 0.6 \text{ MPa}$  ( $\mu = 200 \text{ kPa}$ ). The last bare row is the ballooning limit. For another neo-Hookean silicone, rescale the pressure by  $\mu$ .



**Figure 6:** Deformed section in the bending plane, coloured by the largest principal stretch. Top: bare actuator, with the thin wall ballooning near the limit. Bottom: sleeved actuator.

1. **Bare actuator.** Below  $P/\mu \approx 0.2$  the beam model over-predicts by a factor of 4 to 10. The curves cross near  $P/\mu = 0.43$ ; that is a coincidence of two different mechanisms, not a validation. The response then steepens without bound: between 100 kPa and the limit, 4.6 kPa adds 23 mm of deflection. At the limit the cavity holds four times its original volume and the thin wall is at a third of its thickness.
2. **Hoop-reinforced actuator.** The response is smooth and nearly linear, with no limit point up to 110 kPa. Eq. (5.2) holds within 5% up to  $P/E \approx 0.03$ . At higher pressure the FE angle is larger (by 25% at  $P/E = 0.1$ ) because the cavity grows as the actuator lengthens inside a sleeve of fixed circumference, and the thrust grows with it.
3. **Design consequence.** Reinforce the actuator if a predictable, nearly linear  $P \rightarrow \delta$  map is wanted (e.g. for control). Keep it bare only to study ballooning, and then operate below about 90% of the measured limit pressure.

## 10 Summary of corrections

| Where                         | In the notes                                           | Corrected                                                                                       | Reason                                                  |
|-------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| p. 1, eq. 2                   | $\bar{y} = -r^2 r_p / (R^2 - r)$                       | $\bar{y} = -r^2 r_p / (R^2 - r^2)$                                                              | transcription slip; dimensions                          |
| p. 2, eq. 4                   | $F = PA$                                               | $F = PA_c, A_c = \pi r^2$                                                                       | $A$ already denotes the elastomer area                  |
| p. 3, eq. 7                   | $E = \text{Young's or secant modulus}$                 | small-strain $E = 3\mu = 6C_{10}$                                                               | nonlinearity is ballooning, not a softer $E$            |
| p. 3, $\theta$                | $\theta = \pi P^2 R^2 r_p l / \dots$                   | $\theta = \pi P R^2 r_p l / \dots$                                                              | $\theta = \kappa l$ ; units                             |
| pp. 2–4, eqs. 4–7             | valid for the actuator in general                      | valid with hoop reinforcement; bare: $\times(1 - 2\nu) \approx 0$                               | in-plane stresses cancel the thrust moment              |
| p. 4, note (a)                | strain-dependent tangent modulus in the beam formula   | hyperelastic law in FE/Tier 3                                                                   | bare response is a ballooning effect                    |
| p. 4, note (b)                | $\Delta l = Fl/(EA)$ , shifts the NA                   | bare: $(1 - 2\nu)Fl/(EA) \approx 0$ ; no NA shift                                               | incompressibility; uniform strain                       |
| p. 5, $t(\phi)$               | $\sqrt{R^2 - r_p^2 \sin^2 \theta} - r_p \cos \phi - r$ | $\sqrt{R^2 - r_p^2 \sin^2 \phi} - r_p \cos \phi - r, \phi \text{ from } +y$                     | one angle; exact                                        |
| pp. 5–6                       | $\varepsilon_{zz}, \sigma_z$ longitudinal              | $\varepsilon_x, \sigma_x$                                                                       | axis is $x$ in the page-1 frame                         |
| p. 6, $\kappa_{\text{total}}$ | $+\frac{1}{R+r_p-r}$                                   | $-\frac{1}{R+r_p-r}$                                                                            | consistent with the definition of $\kappa_{\text{exp}}$ |
| p. 6, $\kappa_{\text{exp}}$   | linear in $P$ , superposed on $\kappa_{\text{thrust}}$ | $\varepsilon_{\text{axial}} \approx \frac{2}{3}\varepsilon_{\text{hoop}}^2$ ; FE for prediction | linear term is zero for $\nu = 0.5$                     |
| p. 7, note                    | wall compliance dominates                              | the only first-order mechanism; ends in a limit pressure                                        | FE                                                      |

**Table 6:** Corrections to the handwritten notes.

## 11 Conclusions and next steps

1. The cross-section and loading derivation of the notes (eqs. 1–6) is correct apart from the slips listed above.
2. The Euler–Bernoulli result, eq. (5.2), is the correct first-order model for a *hoop-reinforced* actuator and has been confirmed by FE.
3. For a *bare* silicone actuator the first-order bending is zero, by the factor  $(1 - 2\nu)$ . What bending remains comes from second-order wall lengthening,  $\varepsilon_{\text{axial}} \approx \frac{2}{3}\varepsilon_{\text{hoop}}^2$ , and ends at a ballooning limit  $P_{\text{lim}} \approx 0.52\mu$ .
4. Next: fix the real geometry and silicone, measure the Yeoh constants (ASTM D412 dumbbells), choose bare or reinforced, re-run the FE sweep with the measured material, then fabricate and test against it.

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